A527

Octa-core 64-bit Platform Processor

Overview

A527 series features high-performance platform processors for ARM PCs, such as 2-in-1 laptops, notebooks, virtual desktops, and open-source handheld game consoles. It integrates octa-core Cortex™-A55 CPU, RISC-V, and G57 MC01 GPU to ensure rapid response and smooth running for daily application, such as on-line video, web browsing, and 3D game. A527 series also supports memory interfaces (DDR3/DDR3L/LPDDR3//DDR4/LPDDR4/LPDDR4X, eMMC, NAND, and SPI NAND), high-speed interfaces (PCIe2.1 and USB3.1), multi video output interfaces (2xRGB/2xDual-LVDS/MIPI-DSI/eDP/HDMI), and video input interfaces (MIPI CSI). In addition, this chip family supports 4K@60fps H.265 decoder, 4K@25fps H.264 encoder, DI, and AWonder system to provide users with outstanding video experience. A527 series can be applied in the GMS tablets, the dual screen cash register and the large screen display market as well.

Device Differences

Orderable Devices	Maximum video decoding rate		Package
	H.265	VP9	
A527H000000H	4K@60fps	4K@60fps	FCCSP 660 balls, 17 mm x 17 mm
A527M00X000H	4K@30fps	4K@30fps	ED-FCCSP 660 balls, 17 mm x 17 mm
A527M000000H	4K@30fps	4K@30fps	FCCSP 660 balls, 17 mm x 17 mm

Highlight

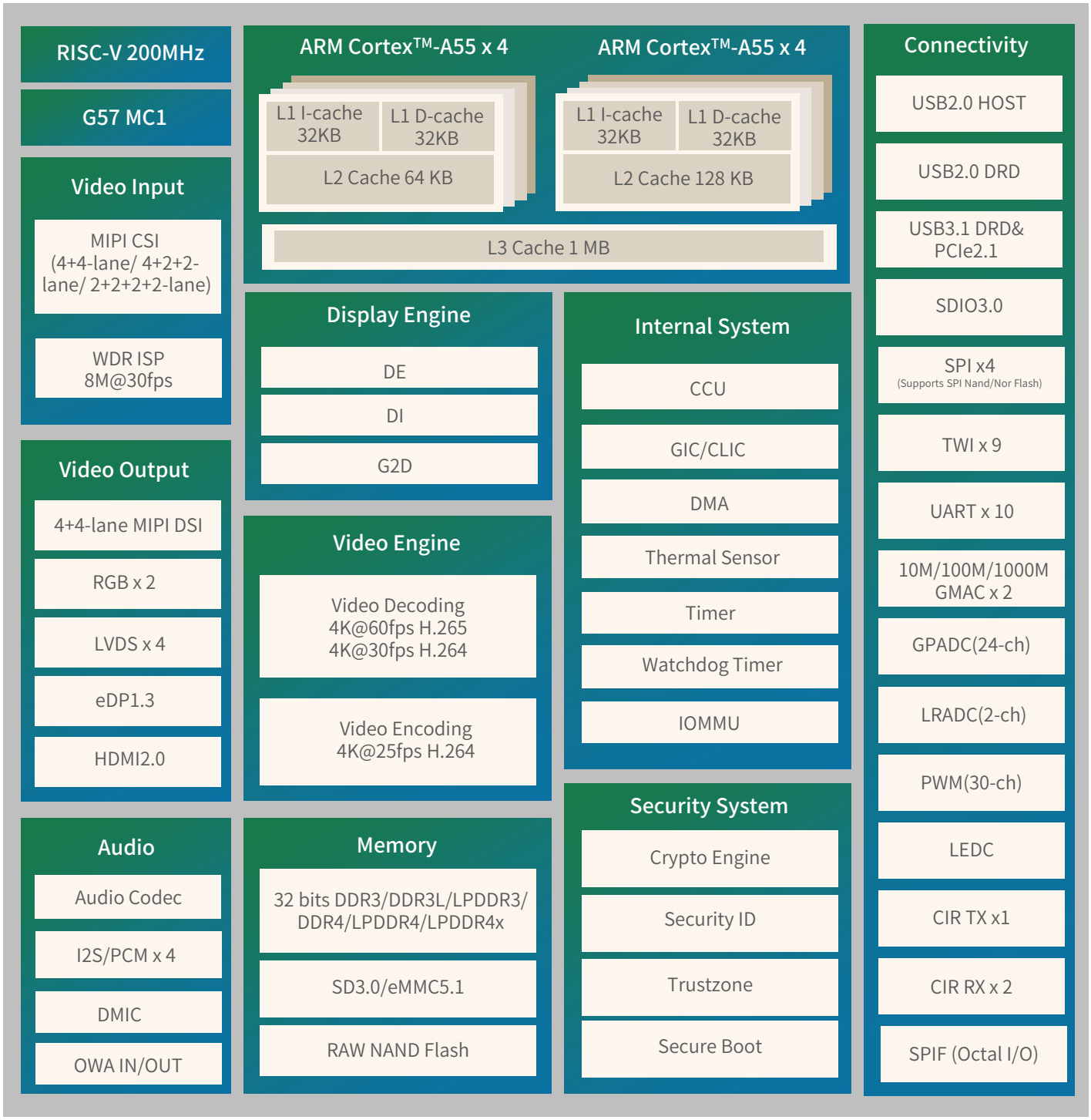
- High performance: A527 series integrates 8 x A55 CPU, RISC-V, G57 MC1 GPU, and 4GB DDR3/DDR3L/LPDDR3/DDR4/LPDDR4/LPDDR4X, which could excellently meet the processing needs of many ARM PC applications.
- Video multimedia: Supports full format video playback, 4K@60fps H.265 video decoder, and 4K@25fps H.264 video encoder.
- Video output: Supports multi video output interfaces such as RGB888, MIPI-DSI, 2x dual-link LVDS, HDMI2.0, and eDP to achieve different display in dual screen.
- Video input: Integrated MIPI CSI can support maximum 4-ch HD camera inputs to achieve differentiated visual functions.
- Rich interfaces: 2 x USB 2.0, 1 x PCIE 2.1&USB 3.1 combo, 2 x GMAC, 10 x UART, 9 x TWI, 30-ch PWM, and 24-ch GPADC greatly facilitate product expansion.

Features

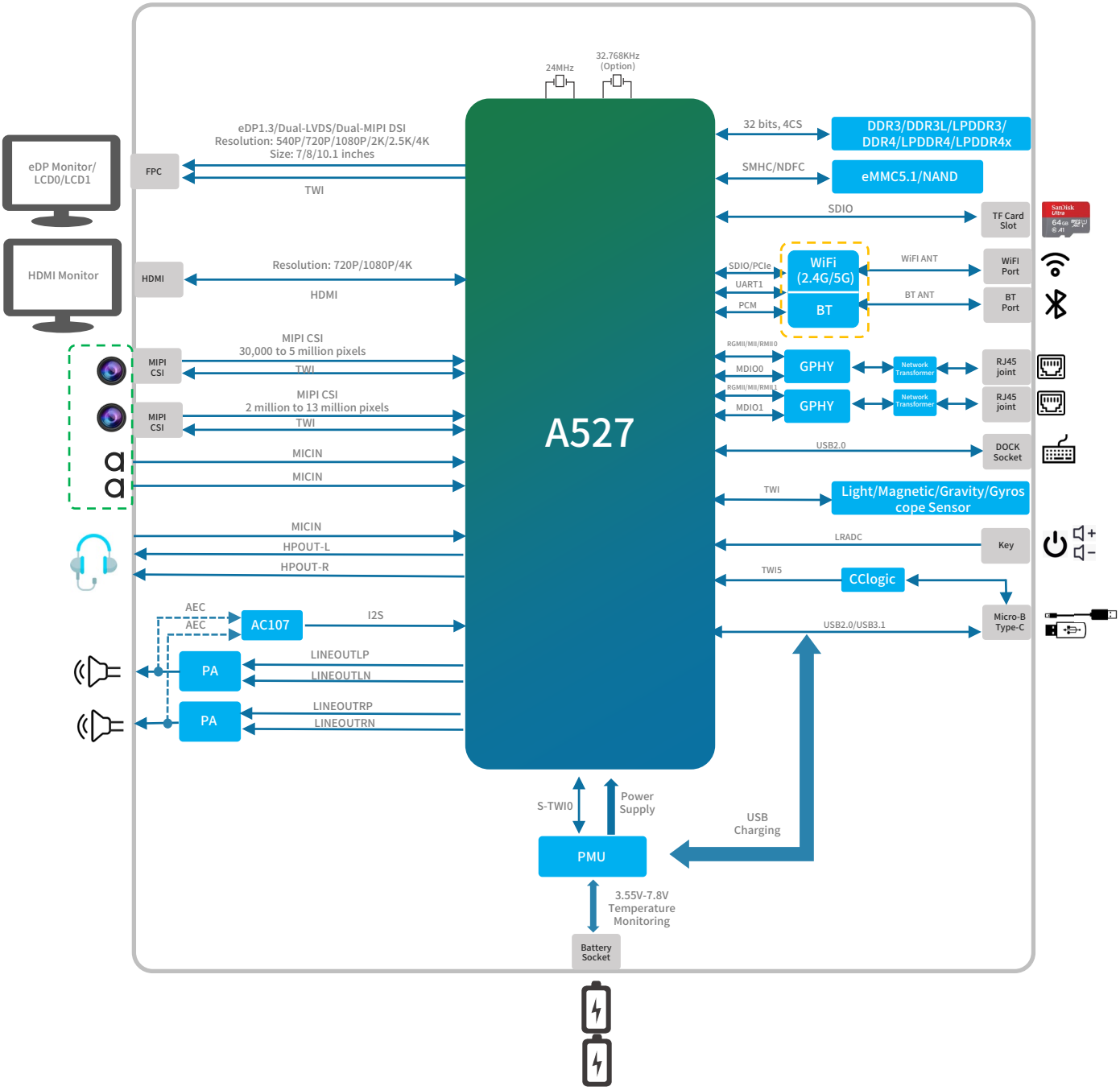
Processor	• Octa-core ARM Cortex™-A55, up to 2.0 GHz • Single-core RISC-V, up to 200 MHz • ARM G57 MC1 GPU
Memory	• 32-bit DDR3/DDR3L/LPDDR3/DDR4/LPDDR4/LPDDR4X interface, supporting maximum capacity of 4GB • 4 chip select lines for LPDDR3, LPDDR4, and LPDDR4X (especially the 64-bit LPDDR3, LPDDR4, and LPDDR4X) • SD3.0/eMMC5.1 interface • 8-bit Nand flash interface with maximum 80-bit/1KB ECC
Video Engine	Video decoder • H.265 MP decoder, up to 4K@60fps • H.264 BL/MP/HP decoder, up to 4K@30fps • VP9 decoder, up to 4K@60fps • Multi-format 1080p@60fps video playback, including VP8, MPEG1/2 SP/MP, MPEG4 SP/ASP, AVS+/AVS JIZHUN Video encoder • H.264 encoder, up to 4K@25fps • MJPEG encoder, up to 4K@15fps • JPEG encoder, up to 8K x 8K resolution
Video Input	• 4+4-lane, 2+2+2+2-lane, or 4+2+2-lane MIPI CSI, up to 2.0 Gbit/s per lane in HS transmission, compliant with MIPI-CSI2 V1.1 and MIPI DPHY V1.1 • Maximum video capture resolution of 8M@30fps • Supports formats: YUV422, YUV420, RAW-8, RAW-10, RAW-12
Video Output	• 4+ 4-lane MIPI-DSI output interface up to 2.5K@60fps and 4K@45fps • Two dual-link LVDS interfaces, up to 1080p@60fps • Two RGB interfaces with DE/SYNC mode, up to 1080p@60fps • eDP 1.3 up to 2.5K@60fps and 4K@30fps • HDMI 2.0b up to 4K@60fps
Audio	• 2 DACs and 3 ADCs • 3 x audio outputs: LINEOUTLP/N, LINEOUTRP/N, HPOUTL/R • 3 x audio inputs: MICIN1P/N, MICIN2P/N, MICIN3P/N • Embedded 4 I2S/PCM interfaces, supporting maximum 16 channels, 8 kHz-384kHz sample rate, and 8-bit-32-bit width • Maximum 8 digital PDM microphones (DMIC) • One OWA input and one OWA output
Security Engine	• Supports Full Disk Encryption • AES, DES, 3DES, and SM4 encryption and decryption algorithms • MD5, SHA, and HMAC tamper proofing • RSA, ECC signature and verification algorithms • Supports 160-bit hardware pseudo random number generator(PRNG) with 175-bit seed • Supports 256-bit hardware true random number generator(TRNG) • Integrated 4K-bit EFUSE for chip ID and security application
Connectivity	• 1 x USB2.0 Host, 1 x USB2.0 DRD, 1 x PCIe2.1&USB3.1 DRD Combo • 2 x GMAC (10/100/1000 Mbit/s port with RGMII and RMII interfaces) • 2 x CIR RX, 1 x CIR TX, 9 x TWI, 4 x SPI, 10 x UART • 30-ch PWM, 24-ch GPADC, 2-ch LRADC • SDIO 3.0, LEDC, SPIF (Octal I/O)

PMIC	• Companion Allwinner Power Management IC
Package	• FCCSP/ED-FCCSP 660 balls • 17 mm x 17 mm size, 0.5 mm ball pitch, 0.3 mm ball size
Process	• 22nm HPC

Block Diagram



Application Diagram



ABOUT ALLWINNER

Allwinner Technology, founded in 2007, is a outstanding designer dedicated to intelligent application SoC, high performance analog component and wireless connectivity IC. It is headquartered in Zhuhai China, with other R&D centers and offices in Shenzhen, HongKong, Xi'an, Beijing and Shanghai. Listed on the GEM of the Shenzhen Stock Exchange in 2015, with the stock code 300458.

Motivated by customer-oriented strategy, Allwinner aligns remarkable R&D teams with long-term core-technology investment in UHD video processing, high-performance multi-core CPU/GPU integration with AI and advanced manufacturing process in terms of high integration , ultra-low power consumption and full-stack integration platform, providing competitive turnkey solutions with considerate services. The products powered by Allwinner spread across from smart hardware, smart home, consumer electronics, HD media, smart video, connected car, industry control, wireless communication to analog products.

CONTACT US

Email：service@allwinnertech.com

This brief is for reference only and has no commitment. All content contained herein is subject to changes without notice.
©2024 Allwinner Technology Co., Ltd.